

Foundation of Ethical Hacking

1. Introduction to Ethical Hacking

- Definition of Ethical Hacking
- Objectives of Ethical Hacking
- Importance in Cybersecurity
- Benefits of Ethical Hacking

2. Define a Hacker

A hacker is a person with advanced technical knowledge who identifies and interacts with vulnerabilities in computer systems.

Depending on their intentions and authorization, hackers are classified into different categories

White Hat Hackers:

- Work legally with permission
- Improve cybersecurity
- Report vulnerabilities responsibly.

Black Hat Hackers

- Gain unauthorized access

- Steal information or damage systems.
- perform illegal activities

Gray Hat Hackers.

- May discover vulnerabilities without authorization
- Usually do not intend harm
- Their actions may still violate laws.

4. Ethics of Ethical Hacking:

- obtain proper authorization.
- Respect privacy and confidentiality.
- Avoid damaging systems.
- Report vulnerabilities honestly
- follow responsible disclosure
- Comply with laws and organizational policies

5. Ethical Hacking & The Legal System

Ethical hackers must operate within legal boundaries by obtaining permission before testing system.

Unauthorized access is illegal in most countries. Ethical hackers should respect contracts, confidentiality agreements, and cybersecurity regulations.

6. proper and Responsible disclosure

Responsible disclosure involves privately reporting vulnerabilities to vendors before making them public.

Steps:

1. identify the vulnerability
2. verify and document the issue.
3. Report it privately.
4. Allow time for a security patch.

7. case study.

Scenario:

A rival security researcher publicly discloses a vulnerability and takes full credit, even though you had already privately reported it to the vendor under an embargo.

8. Ethical norms Around disclosure credit

Researchers who responsibly discover vulnerabilities deserve proper acknowledgment. If multiple researchers independently discover the same vulnerability, vendors should recognize all legitimate

contributors whenever possible.

(9) Importance of embargoes

An embargo is a confidential period during which vulnerability details remain private until a security update is available.

Benefits:

- protects users.
- Gives vendors time to develop patches
- supports coordinated disclosure.

(10) Handling credit disputes professionally

Researches should:

- Remain professional and respectful
- contact the vendor privately.
- Respect appropriate acknowledgment.
- Avoid public arguments or accusations
- Request appropriate acknowledgment

(11) conclusion,

Ethical hacking is based on responsibility, integrity, legality and professionalism. Responsible disclosure and embargoes protect users while enabling vendors to fix vulnerabilities.

Although receiving proper recognition is important, researches should handle disputes respectfully.